

## **Mission Sentence**

For patent attorneys who need help assessing whether a draft claim is novel, “Patent Intelligence” is a feature-level prior-art evidence retriever that breaks a claim in its individual features and returns, for each feature, the specific passages of prior art patents that disclose it. Unlike current patent attorney search tools, such as keyword searches over whole documents, it works at the level of individual claim features and shows the source passage behind every match, so that the attorney can accept or reject a hit without opening a reference.

## **Named User**

- *Patent Attorneys/Agents (Primary)* - Assess a client’s draft claim before filing. Need help finding the relevant patent for determining novelty of an application. Low risk, have an expert “in the loop” and can verify outputs.
- *Existing Companies* - Need to determine whether a competitor’s newly patented product infringes their current patents (medium risk, needs precision but has resources to pay for human experts and correct any problems)
- *Startups, Inventors, & Researchers* - Need to know whether their product, invention, or research field will be granted a patent, or whether there are relevant patents that will prevent this (high risk, needs precision but lacks the resources to correct any problems)
- *United States Patent & Trademark Office Admin* - Wants to determine whether patent application should be granted in a quick, accurate, and cheap manner, and needs to find relevant patents (high risk, needs precision and any mistakes could clog the legal system)

## Top-5 Stories with Acceptance Criteria

- *(Primary User)* As a patent attorney, I want to quickly find relevant patents, so that I can quickly and efficiently determine whether my client's patent application is novel.
  - Acceptance: Relevant patents are found in under 1 minute and there is a 75% overlap with what the attorney would have used.
- *(Primary User)* As a patent agent, I want to quickly find relevant patents, so that I can inform my client of nonnovel patent applications before money and time is wasted on the legal process.
  - Acceptance: Relevant patents are found in under 1 minute and there is a 75% overlap with what the agent would have referenced.
- As an inventor, I want to determine whether my work/idea is novel before applying for a patent, so that I can save money on legal fees and spend more time on potentially novel and lucrative ideas.
  - Acceptance: Relevant patents are found in under 10 minutes and align with what a lawyer would use 90% of the time.
- As a researcher, I want to determine whether my area of research is novel, so that I only spend time on novel ideas and do not waste grants and other research funding.
  - Acceptance: Relevant patents found in under 10 minutes, and continue to be found as new, relevant patents are granted, and retrieved patents are relevant 75% of the time.
- As a United States Patent and Trademark Office administrator, I want to determine whether an applicant's patent is novel without dealing with lots of reviewers, so that I have a cheaper, more consistent, and faster system.

- Acceptance: Relevant patents are retrieved in under 1 minute and aligns with what a USPTO employee would retrieve 99% of the time.

## **Other Stories**

- As a lawyer for a large company, I want to determine whether there are patents similar to my client's patent, so that I can defend their intellectual property and take legal action.
  - Acceptance: Generates newer patents potentially relevant to the old patent in under 10 minutes and it aligns with what the lawyer would say 90% of the time.

## Assumption Table with Test Results

| Assumption                                                                                  | Kill Criteria                                                                                                | Test                                                             | Result (Cannot Talk to Users Yet)                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An AI model can efficiently search through a patent database and find the relevant patents. | An AI model cannot be run efficiently or updated easily.                                                     | Look at what past research papers have done in the field.        | Inconclusive but not good. Previous papers used a <a href="#">multibillion parameter model</a> . Since the model would need to be consistently updated as new patents are granted, the model would have to run again, so this might be a problem. |
| The use of AI tools for determining patent novelty is legal and unregulated.                | There are excessive regulations against the use of AI when determining novelty and in the invention process. | Research relevant laws, judicial rulings, and executive orders.  | According to <a href="#">current law</a> , “AI may be used as a tool, but human beings must conceive the invention.” The USPTO <a href="#">is allowed to use AI</a> in research.                                                                  |
| USPTO Data is free to use, easily viewable, and downloadable.                               | USPTO data is not usable and there are no suitable replacements.                                             | Look up the USPTO database and try to find and download patents. | Found the <a href="#">relevant database</a> , and also found <a href="#">Google Patents</a> , which might be easier to download and manage.                                                                                                       |

## Claude Opus 5 Generated Assumptions

1. An attorney will act on feature-level evidence without a novelty verdict.
  - a. Why fatal: The entire product thesis. If attorneys want a yes/no answer, this tool has no user.
  - b. Kill criterion: Pivot if fewer than 2 of 3 attorneys say evidence-without-verdict would change how they search, or if any says they would not use a tool that declines to give a conclusion.
  - c. Test: Interview 3 practitioners — a BU Technology Development licensing officer, a patent attorney or agent, and a third from the BU alumni network.
  - d. Result: **NOT YET RUN**. Cheapest test available and the one that most needs doing. No code should be written before it.

- e. Note: Not originally considered. A tool that the user does not use is pointless
2. Draft claims can lawfully and ethically be sent to the system.
- a. Why fatal: Unfiled draft claims are confidential and unpublished. If they cannot be transmitted to a third-party API, the architecture must be local-only, which changes every downstream decision.
  - b. Kill criterion: Pivot to open-weight, locally hosted models if any of the three interviewed practitioners says firm policy or professional obligation bars sending unfiled claim text to a third-party service.
  - c. Test: Review USPTO guidance and ABA Model Rule 1.6 commentary on cloud services; ask the same three practitioners directly.
  - d. Result: **PARTIAL**. Prior work is suggestive: the FiNE-Patents authors restricted their analysis to open-weight models specifically because patent examination demands privacy and security standards that are difficult to guarantee with closed-weight models accessed via API [3]. Direct guidance not yet located; practitioner question not yet asked.
  - e. Note: Not originally considered. Need to further research the regulations around this and third part viewing.
3. This beats what the attorney does today
- a. Why fatal: free tools already exist. Google Patents offers a similar-documents feature; commercial platforms include PatSnap, Derwent Innovation, Questel Orbit, and LexisNexis PatentAdvisor. If the incumbent already surfaces the examiner-cited art, there is no project.
  - b. Kill criterion: Pivot if Google Patents' similar-documents feature already surfaces

the examiner-cited reference in its top 10 for 8 or more of 10 test claims.

- c. Test: One hour. Take 10 claims from FiNE-Patents whose examiner-cited prior art is known, run each through Google Patents, and record whether the cited reference appears in the top 10 — and whether the tool identifies which passage matters.
  - d. Result: **NOT YET RUN**.
  - e. Note: This was not originally considered, but will be a key part of testing
4. Feature-level retrieval is achievable at student scale
- a. Why fatal: Determines whether the project is buildable on available compute.
  - b. Kill criterion: Pivot to a smaller CPC subclass or a lexical-plus-reranker architecture if a 30B-class open-weight model cannot process one claim against one prior-art document in under 60 seconds on available hardware.
  - c. Test: Read the reported configurations and results in the three papers from Phase 1.
  - d. Result: **PASS**, with a correction to an earlier draft. A 30B open-weight model using hierarchical (feature-by-feature) examination reached 53.7 F1 on novel feature identification, beating a 397B model using single-step examination at 44.7 F1 — structured decomposition mattered more than model scale for that sub-task [3]. A 30B-class model is within reach. The 397B model is a mixture-of-experts with roughly 17B active parameters, not a dense 400B model, so its inference cost was previously overstated.
  - e. Note: Claude seems to think I can run the model, but I have my doubts.

5. USPTO and EPO data are free, accessible, and downloadable
- a. Why least fatal: several independent sources exist, so failure of any one is recoverable.
  - b. Kill criterion: Pivot to the pre-packaged FiNE-Patents release if bulk download and parsing of raw patent data takes more than two days.
  - c. Test: Locate and download patent full texts and metadata from each candidate source.
  - d. Result: **PASS**. USPTO bulk data is publicly available at [bulkdata.uspto.gov](https://bulkdata.uspto.gov), EPO full texts via the Publication Server API, and Google Patents offers an easier-to-manage interface for spot checks. The FiNE-Patents dataset and creation pipeline are released publicly at [github.com/boschresearch/fine-patents](https://github.com/boschresearch/fine-patents), which provides a fallback that removes this risk almost entirely.
  - e. Note: Considered and passed.

[Link to Claude Chat](#)